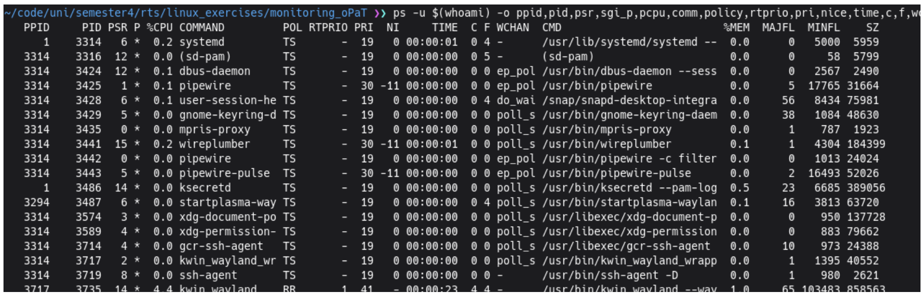
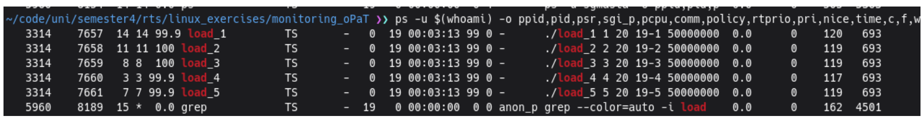
Monitoring of Processes and Threads

Exercises 1

a)





Flag	Meaning
-u	user
-o	Display the following columns
PPID	parent process id
PID	process id
PSR	designatedCPU core number
SGI_P	cpu core its actually running on rn
%CPU	cpu core usage in %
COMM	full command without args
POLICY	scheduling policy
RTPRIO	Real time priority
PRI	Prio of the process
NICE	niceness
TIME	used cpu time
C	full cpu usage value
F	process flags (for kernel)
WCHAN	what sleep function from the kernel the process is using
CMD	full command with args
%MEM	memory usage in %
MAJFLT	major page faults
MINFLT	minor page faults
SZ	size of process given in pages (4kb)

b)

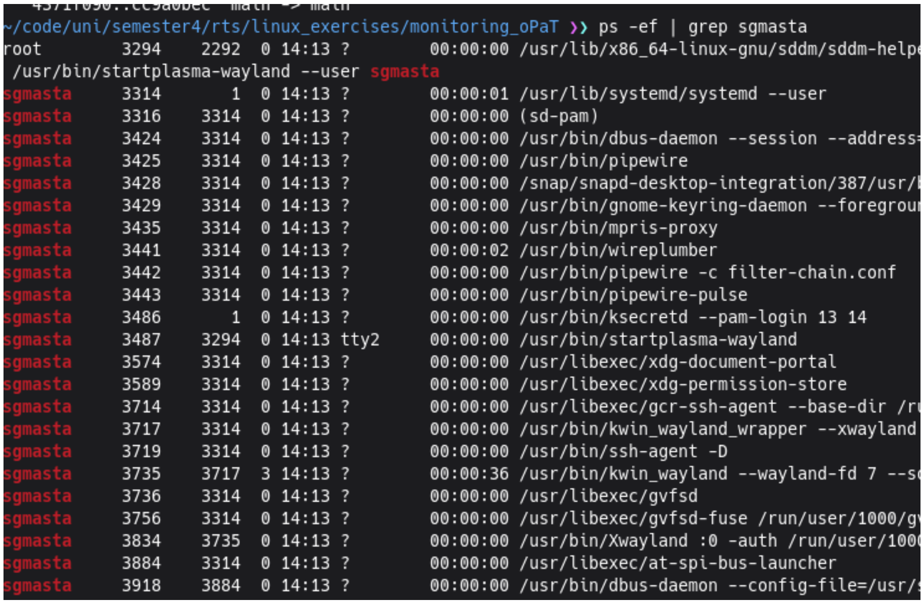
```
~/code/uni/semester4/rts/linux_exercises/monitoring_oPaT ❯❯  pstree -acghlpsUu
systemd,1,1 --switched-root --system --deserialize=52 splash
  ├─ModemManager,1861,1861
  │   ├─{ModemManager},1869,1861
  │   ├─{ModemManager},1870,1861
  │   └─{ModemManager},1873,1861
  ├─NetworkManager,1755,1755 --no-daemon
  │   ├─{NetworkManager},1855,1755
  │   ├─{NetworkManager},1857,1755
  │   └─{NetworkManager},1858,1755
  ├─accounts-daemon,1657,1657
  │   ├─{accounts-daemon},1718,1657
  │   ├─{accounts-daemon},1719,1657
  │   └─{accounts-daemon},1749,1657
  ├─avahi-daemon,1619,1619,avahi
  │   └─avahi-daemon,1682,1619
  ├─bluetoothd,1620,1620
  ├─chronyd-starter,1622,1622 /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
  │   └─chronyd,1743,1622,_chrony -n -F 1
  │       └─chronyd,1754,1622 -n -F 1
  ├─colord,3278,3278,colord
  │   ├─{colord},3282,3278
  │   ├─{colord},3283,3278
  │   └─{colord},3285,3278
  ├─containerd,1971,1971
  │   ├─{containerd},2021,1971
```

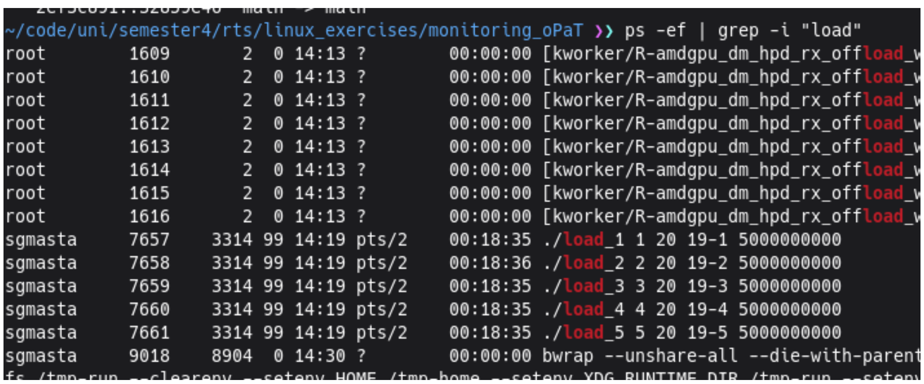
```
~/code/uni/semester4/rts/linux_exercises/monitoring_oPaT ❯❯ pstree -acghlpsUu | grep -i "load"
 │    ├─load_1,7657,7647 1 20 19-1 5000000000
 │    ├─load_2,7658,7647 2 20 19-2 5000000000
 │    ├─load_3,7659,7647 3 20 19-3 5000000000
 │    ├─load_4,7660,7647 4 20 19-4 5000000000
 │    ├─load_5,7661,7647 5 20 19-5 5000000000
 │    │    │    │    ├─grep,8503,8502 --color=auto -i load
```

Flag	Meaning
-a	Shows command line args
-c	doesnt truncate branches, you see everything
-g	shows group ids
-h	highlights parent process
-l	doesnt truncate long strings
-p	shosws pids
-s	shows process inheritance
-U	uses unicode
-u	shows different users

c)

The command `ps -ef` shows all processes detailed





Minor page fault

A minor page fault happens when the program tries to acces a page that is not currently loaded into its virtual address space. When the page is outside this address space but still insdie the RAM, a minor page fault occurs, which means that the system need to update the page table.

This only costs little time and effort compared to a major page fault, where the requested page is not only outside the virtual address space, but also outside the RAM (on a harddisk)

This means that the system needs to swap from the harddrive, which is very time and resource expensive